

Toward a Dynamical Theory of Emotion in Psychophysical Interaction Systems: Second-Order Dynamics and Relaxation of Time

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Abstract

We extend the continuous-time interaction framework introduced in A Continuous Dynamical Systems Framework for Interpersonal Interaction (Pinmook, 2025) [3] [4] by deriving a second-order dynamical equation governing emotional acceleration in interpersonal systems. Using the generator matrix $\mathbf{M}(t)$ associated with interpersonal psychophysical flow, we prove that emotional quantity emerges as a second-order response term of interaction, $\mathbf{E}(t) = \dot{\mathbf{M}}(t) + \mathbf{M}(t)^2$. The resulting equation,

$$\ddot{\mathbf{I}}_S(t) = \mathbf{E}(t)\mathbf{I}_S(t),$$

defines emotion as a mathematically necessary property of time-varying perception-action coupling. We additionally derive the necessary and sufficient condition under which interaction “relaxes” toward emotional equilibrium, proving that $M(t) = (t + \tau_0)^{-1}$ is the unique class of generator functions yielding zero emotional acceleration. This provides the first mathematically rigorous demonstration that emotional healing corresponds to a monotonic decay of interaction flow over time, resembling meditative emotional detachment. Our analysis solidifies the psychophysical interaction program as a three-layer hierarchy: discrete dynamics, continuous flow, and emotional acceleration.

1 Mathematical Preliminaries and Revisited Results

We begin by recalling the continuous interaction equation proved in [Pinmook 2025]:

Theorem 1 (Continuous Interaction Equation). Let $\mathbf{I}_S(t) \in \mathbb{R}^n$ denote the active interaction state of an n -agent system. Then

$$\frac{d\mathbf{I}_S}{dt} = \mathbf{M}(t)\mathbf{I}_S(t),$$

where $\mathbf{M}(t) \in \mathbb{R}^{n \times n}$ is the interaction flow generator satisfying

$$\gamma = \exp(\mathbf{M}\Delta t),$$

for discrete projection matrix γ .

We assume $\mathbf{M}(t)$ be a time-dependent diagonal matrix of the form

$$\mathbf{M}(t) = \text{diag}(m_{11}(t), m_{22}(t), \dots, m_{nn}(t))$$

where each $m_{ii}(t) \in C^1(\mathbb{R}^+)$.

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2 Derivation of the Second-Order Emotional Equation

We compute the second derivative of $\mathbf{I}_S(t)$:

$$\begin{aligned}\frac{d}{dt} \left(\frac{d\mathbf{I}_S}{dt} \right) &= \frac{d}{dt} (\mathbf{M}(t)\mathbf{I}_S(t)) \\ &= \dot{\mathbf{M}}(t)\mathbf{I}_S(t) + \mathbf{M}(t)\frac{d\mathbf{I}_S(t)}{dt}.\end{aligned}$$

Using Theorem 1, $\dot{\mathbf{I}}_S = \mathbf{M}\mathbf{I}_S$:

$$\begin{aligned}\ddot{\mathbf{I}}_S &= \dot{\mathbf{M}}\mathbf{I}_S + \mathbf{M}(t)^2\mathbf{I}_S \\ &= (\dot{\mathbf{M}}(t) + \mathbf{M}(t)^2)\mathbf{I}_S\end{aligned}$$

Definition 1.1 (Emotion Quantity).

We define the emotion observable matrix:

$$\mathbf{E}(t) := \dot{\mathbf{M}}(t) + \mathbf{M}(t)^2.$$

It is the interaction curvature operator induced by the time-variation of the interaction generator $\mathbf{M}(t)$.

Thus the emotional acceleration equation becomes:

Theorem 2 (Equation of Emotion).

$$\ddot{\mathbf{I}}_S(t) = \mathbf{E}(t)\mathbf{I}_S(t). \quad (1)$$

Emotion is the second-order curvature of interaction flow.

Proposition 1 (Componentwise Decoupling)

If $\mathbf{M}(t)$ is diagonal, then the emotional relaxation condition

$$\mathbf{E}(t) = \mathbf{0}$$

is equivalent to the system of independent scalar Riccati equations

$$\dot{m}_{ii}(t) + m_{ii}(t)^2 = 0, \quad i = 1, \dots, n.$$

Proof : Since $\mathbf{M}(t)$ is diagonal,

$$\dot{\mathbf{M}}(t) = \text{diag}(\dot{m}_{11}(t), \dot{m}_{22}(t), \dots, \dot{m}_{nn}(t),), \quad \mathbf{M}(t)^2 = \text{diag}(m_{11}(t)^2, m_{22}(t)^2, \dots, m_{nn}(t)^2).$$

Therefore,

$$\mathbf{E}(t) = \dot{\mathbf{M}}(t) + \mathbf{M}(t)^2 = \text{diag}(\dot{m}_{11} + m_{11}^2, \dots, \dot{m}_{nn} + m_{nn}^2)$$

Thus $\mathbf{E}(t) = \mathbf{0}$ if and only if each diagonal component vanishes:

$$\dot{m}_{ii}(t) + m_{ii}(t)^2 = 0, \quad \forall i.$$

Theorem 3 (Unique Relaxation Solution for Diagonal Generators) For each $i = 1, \dots, n$, the unique maximal solution of

$$\dot{m}_{ii}(t) = -m_{ii}(t)^2$$

with initial condition $m_{ii}(0) = m_{i,0} \neq 0$ is

$$m_{ii}(t) = \frac{1}{t + \tau_i}, \quad \tau_i := \frac{1}{m_{i,0}}.$$

Proof : The equation is separable:

$$\frac{dm_{ii}}{m_{ii}^2} = -dt.$$

Integrating:

$$-\frac{1}{m_{ii}(t)} = -t + C_i,$$

where C_i be a constant.

Solving for $m_{ii}(t)$:

$$m_{ii}(t) = \frac{1}{t + \tau_i}, \quad \tau_i := C_i.$$

Applying the initial condition:

$$m_{ii}(0) = \frac{1}{\tau_i} = m_{i,0} \Rightarrow \tau_i = \frac{1}{m_{i,0}}.$$

Such that τ_i is an intrinsic relaxation scale of generator $\mathbf{M}(t)$.

Corollary 1 (Diagonal Emotional Relaxation Matrix)

Under emotional relaxation,

$$\mathbf{M}(t) = \text{diag} \left(\frac{1}{t + \tau_1}, \frac{1}{t + \tau_2}, \dots, \frac{1}{t + \tau_n} \right).$$

Each diagonal entry decays monotonically:

$$\lim_{t \rightarrow \infty} m_{ii}(t) = 0.$$

Thus:

$$\lim_{t \rightarrow \infty} \mathbf{M}(t) = 0_{n \times n}.$$

Theorem 4 (Interaction-Level Consequence) Let the interaction dynamics satisfy

$$\dot{\mathbf{I}}_S(t) = \mathbf{M}(t)\mathbf{I}_S(t),$$

with diagonal relaxation matrix above. Then each component evolves as

$$I_{S,i}(t) = C_i(t + \tau_i),$$

where $C_i \in \mathbb{R}$.

Proof : Componentwise:

$$\dot{I}_{S,i} = m_{ii}(t)I_{S,i} = \frac{1}{t + \tau_i}I_{S,i}.$$

Separating variables:

$$\frac{dI_{S,i}}{I_{S,i}} = \frac{dt}{t + \tau_i}.$$

Integrating:

$$\ln |I_{S,i}(t)| = \ln(t + \tau_i) + C'_i$$

$$I_{S,i}(t) = C_i(t + \tau_i)$$

3 Emotional Relaxation Beyond the Diagonal Case

3.1 Normal Flow Matrices

Definition 2.1 (Normal Matrix)

A matrix $\mathbf{M}$ is normal if

$$\mathbf{M}\mathbf{M}^T = \mathbf{M}^T\mathbf{M}.$$

Normal matrices are unitarily diagonalizable: there exists an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{M} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T,$$

where $\mathbf{\Lambda}$ is diagonal.

Assumption A (Spectral Frame Invariance)

Assume $\mathbf{M}(t)$ is normal for all t and shares a time-independent eigenbasis, i.e.

$$\mathbf{M}(t) = \mathbf{Q}\mathbf{\Lambda}(t)\mathbf{Q}^T, \quad \mathbf{Q}^T\mathbf{Q} = \mathbf{I}.$$

This assumption is minimal and standard in time-dependent spectral theory.

Theorem 5 (Reduction to Diagonal Relaxation)

Under Assumption A, the emotional relaxation condition is equivalent to

$$\dot{\mathbf{\Lambda}}(t) + \mathbf{\Lambda}(t)^2 = \mathbf{0},$$

i.e. to independent scalar equations

$$\dot{\lambda}_i(t) + \lambda_i(t)^2 = 0, \quad i = 1, 2, \dots, n$$

Proof : Since $\mathbf{Q}$ is constant,

$$\dot{\mathbf{M}} = \mathbf{Q}\dot{\mathbf{\Lambda}}\mathbf{Q}^T, \quad \mathbf{M}^2 = \mathbf{Q}\mathbf{\Lambda}^2\mathbf{Q}^T.$$

Thus

$$\mathbf{E}(t) = \dot{\mathbf{M}} + \mathbf{M}^2 = \mathbf{Q}(\dot{\mathbf{\Lambda}} + \mathbf{\Lambda})\mathbf{Q}^T.$$

Because $\mathbf{Q}$ is invertible,

$$\mathbf{E}(t) = \mathbf{0} \Leftrightarrow \dot{\mathbf{\Lambda}} + \mathbf{\Lambda}^2 = \mathbf{0}.$$

Corollary 2.1 (Normal Relaxation Flow)

Each eigenvalue satisfies

$$\lambda_i(t) = \frac{1}{t + \tau_i},$$

and therefore

$$\mathbf{M}(t) = \mathbf{Q} \text{diag} \left(\frac{1}{t + \tau_1}, \dots, \frac{1}{t + \tau_n} \right) \mathbf{Q}^T$$

The matrix norm decays as

$$\|\mathbf{M}(t)\| = \mathcal{O}(t^{-1}).$$

3.2 General Diagonalizable Flow Matrices

We now drop normality.

Definition 3.1 (Diagonalizable Generator)

$\mathbf{M}(t)$ is diagonalizable if

$$\mathbf{M}(t) = \mathbf{P} \mathbf{\Lambda}(t) \mathbf{P}^{-1}$$

for some invertible (not necessarily orthogonal) $\mathbf{P}$.

Theorem 3.1 (Necessary Condition for Relaxation)

Let

$$\mathbf{M}(t) = \mathbf{P} \mathbf{\Lambda}(t) \mathbf{P}^{-1}, \quad \mathbf{P} \text{ constant.}$$

Then emotional relaxation implies

$$\dot{\mathbf{\Lambda}} + \mathbf{\Lambda}^2 = \mathbf{0} \quad \text{and} \quad [\mathbf{P}, \mathbf{\Lambda}(t)] = 0.$$

Proof : We compute

$$\dot{\mathbf{M}} = \mathbf{P} \dot{\mathbf{\Lambda}} \mathbf{P}^{-1}, \quad \mathbf{M}^2 = \mathbf{P} \mathbf{\Lambda}^2 \mathbf{P}^{-1}.$$

Thus

$$\mathbf{E}(t) = \mathbf{P}(\dot{\mathbf{\Lambda}} + \mathbf{\Lambda}^2) \mathbf{P}^{-1}.$$

Hence $\mathbf{E} = \mathbf{0} \Rightarrow \dot{\mathbf{\Lambda}} + \mathbf{\Lambda}^2 = \mathbf{0}$.

However, consistency requires that eigenvectors remain invariant in time. If $\mathbf{P}$ depended on t , additional commutator terms $[\dot{\mathbf{P}} \mathbf{P}^{-1}, \mathbf{\Lambda}]$ would appear, preventing $\mathbf{E} = \mathbf{0}$.

3.3 Invariant Healing Time

Definition : T_{heal} is an invariant healing time if

$$T_{\text{heal}}(\epsilon) := \inf \{T : \|\mathbf{E}(t)\| < \epsilon, \quad \forall t \geq T\}$$

Lemma : Let $\mathbf{M}(t) \in C^1(\mathbb{R}^+, \mathbb{R}^{n \times n})$, if $\mathbf{M}(t) = \frac{1}{t+\tau} + \delta \mathbf{M}(t)$ and $\|\delta \mathbf{M}(t)\| \leq C(t+\tau)^{-2}$, then

$$T_{\text{heal}}(\epsilon) \sim \mathcal{O}(\epsilon^{-1/2}) - \tau.$$

Proof : Finding a derivative of $\mathbf{M}(t)$,

$$\dot{\mathbf{M}}(t) = -\frac{1}{(1+\tau)^2} + \delta \dot{\mathbf{M}}(t).$$

and

$$\mathbf{M}(t)^2 = \left(\frac{1}{t+\tau} + \delta\mathbf{M}(t) \right)^2 = \left(\frac{1}{t+\tau} \right)^2 + 2 \left(\frac{1}{t+\tau} \delta\mathbf{M}(t) \right) + \delta\mathbf{M}(t)^2.$$

From a definition of an emotion quantity,

$$\mathbf{E}(t) = \dot{\mathbf{M}}(t) + \mathbf{M}(t)^2,$$

$$\mathbf{E}(t) = -\frac{1}{(1+\tau)^2} + \delta\dot{\mathbf{M}}(t) + \left(\frac{1}{t+\tau} \right)^2 + 2 \left(\frac{1}{t+\tau} \delta\mathbf{M}(t) \right) + \delta\mathbf{M}(t)^2,$$

$$\mathbf{E}(t) = \delta\dot{\mathbf{M}}(t) + 2 \left(\frac{1}{t+\tau} \delta\mathbf{M}(t) \right) + \delta\mathbf{M}(t)^2.$$

Since $\|\delta\mathbf{M}(t)\| \leq C(t+\tau)^{-2}$, we therefore give that

$$\left\| 2 \frac{1}{t+\tau} \delta\mathbf{M}(t) \right\| \leq 2C(t+\tau)^{-3}.$$

And we give that

$$\|\delta\mathbf{M}(t)^2\| = \|\delta\mathbf{M}(t)\|^2 \leq C^2(t+\tau)^{-4}.$$

By the properties of smoothness C^1 ,

$$\left\| \delta\dot{\mathbf{M}}(t) \right\| \leq C'(t+\tau)^{-3}.$$

So, the emotion quantity can be written as

$$\|\mathbf{E}(t)\| \leq C'(t+\tau)^{-3} + 2C(t+\tau)^{-3} + C^2(t+\tau)^{-4} \leq K(t+\tau)^{-2}.$$

for large t and $K > 0$.

Solving for an invariant healing time, we can assume that

$$\|\mathbf{E}(t)\| < \epsilon, \quad \forall t \geq T,$$

sufficient large t for that

$$K(t+\tau)^{-2} < \epsilon.$$

Solving that,

$$t+\tau > \sqrt{\frac{K}{\epsilon}}.$$

We find that an invariant healing time, $T_{\text{heal}}(\epsilon)$, a boundary of time for relaxation, as

$$T_{\text{heal}}(\epsilon) > \sqrt{\frac{K}{\epsilon}} - \tau.$$

Therefore,

$$T_{\text{heal}}(\epsilon) \sim \mathcal{O}(\epsilon^{-1/2}) - \tau, \quad \text{as } \epsilon \rightarrow 0.$$

3.4 Failure of Relaxation in Non-Diagonalizable Systems

Proposition 4.1 (Jordan Obstruction) If $\mathbf{M}(t)$ contains a nontrivial Jordan block, then

$$\mathbf{E}(t) = \mathbf{0}$$

has no solution.

Sketch of Proof : Let

$$\mathbf{M}(t) = \mathbf{P}(J(t))\mathbf{P}^{-1}, \quad J = \lambda\mathbf{I} + N, \quad N \neq 0, \quad N^2 = 0.$$

Then

$$\mathbf{M}^2 = \mathbf{P}(\lambda^2\mathbf{I} + 2\lambda N)\mathbf{P}^{-1}.$$

Even if $\dot{\lambda} + \lambda^2 = 0$, the nilpotent term $2\lambda N$ survives, so

$$\mathbf{E}(t) \neq \mathbf{0}.$$

4 Discussion

This final paper confirms results from many researches. As Lazarus (1991) noted, emotions are products of continuous interpretation of environmental relationships, a concept that aligns with the proposed mathematical definition of emotion as a second-order response to interaction flow (Lazarus, 1991) [2]. By Klaus Scherer (2001), an emotion is defined as an episode of interrelated, synchronized changes in the states of all or most of the five organismic subsystems in response to the evaluation of an external or internal stimulus event as relevant to major concerns of the organism (Scherer, 2001) [5].

This result also supports a "relaxation" of interaction by a reduction of interaction flow matrix $\mathbf{M}(t) \rightarrow \mathbf{0}$ for a calm event (emotion does not be in an interaction loops). From a conception of Hayes (2011), psychological flexibility is established through... acceptance, cognitive defusion, being present, and self-as-context—to undermine the grip of literal language and allow for a more values-consistent life [1].

completes the structure:

Table 1: Structure of interaction equation

Layer	Equation	Phenomenon
Discrete	$\mathbf{I}_S(t+1) = \gamma\mathbf{I}_S(t)$	turn-based social interaction
Continuous	$\dot{\mathbf{I}} = \mathbf{M}\mathbf{I}_S$	real-time affective flow
Emotional	$\ddot{\mathbf{I}} = \mathbf{E}\mathbf{I}_S$	emotion as acceleration

Emotion is not an add-on — it is mathematically forced by differentiability of interaction flow. This result opens a gateway to:

1. nonlinear emotional turbulence (future work: Riccati-type)
2. stochastic emotional noise (Itô-diffusion on $\mathbf{M}$)
3. empirical parameter-learning from interpersonal recordings

5 Conclusion

We have shown:

$$\mathbf{E}(t) = \dot{\mathbf{M}}(t) + \mathbf{M}(t)^2 \Rightarrow \ddot{\mathbf{I}}_S(t) = \mathbf{E}(t)\mathbf{I}_S(t)$$

and proved the necessary-and-sufficient condition:

$$\mathbf{E}(t) = \mathbf{0} \Leftrightarrow \mathbf{M}(t) = \frac{1}{t + \tau_0}.$$

Thus emotional quantity is mathematically equivalent to decay of perceptual-action flow. It is controlled by a The intrinsic relaxation scale τ , in a unit of time scale, which characterizes the relative rate at which interaction strength decays over time, rather than determining the moment at which emotional equilibrium is reached. Furthermore, we give that an invariant healing time T_{heal} , as a boundary of interaction relaxation.

References

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